Linear Systems on Graphs

Controllability of Structured Networks

Chapter 1

Introduction

Systems and Models · From One Plant to Networks · Book Outline

Department of Control & Instrumentation Engineering
Pukyong National University, Busan, Korea
pnj@pknu.ac.kr

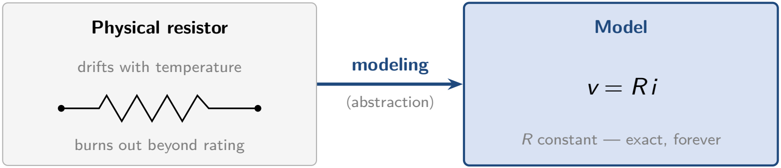
Nam-Jin Park

Chapter 1 Contents

From Physical System to Model

CONCEPT Modeling: Physical System $\rightarrow$ Model

- Engineering **analysis** never touches the physical system — it operates on a **model we build** by modeling
- Resistor: the real part drifts with temperature and burns out beyond rating $\rightarrow$ we model it as a **constant** R
- Robot link: the real link flexes under load $\rightarrow$ we model it as **perfectly rigid**
- Every later computation is bounded by the **quality of the model** — **the model is not reality**



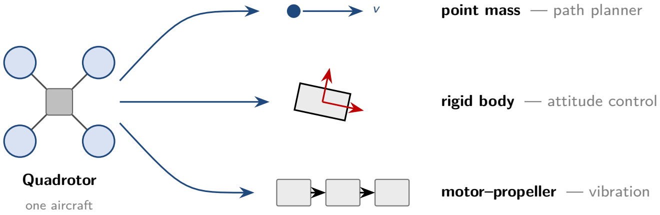
▶ Convention of This Course

- From now on, "system" always means **a model of a physical system**
- Modeling is assumed done — our starting point is the model itself

One Physical System, Three Models

EXAMPLE Quadrotor

- To a path planner: a **point mass**
- To an attitude controller: a **rigid body**
- To a vibration analyst: **coupled motor–propeller dynamics**



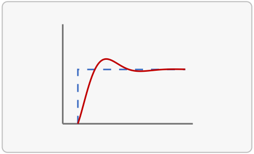
Three models, one aircraft — no contradiction. **The question being asked** decides which model is appropriate.

Four Stages of Analytical Study — Where This Course Lives

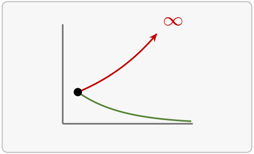
Modeling	→	Description	→	**Analysis**	→	Design
physical system → model		model → mathematical equations (state space)		predict and judge behavior		modify to meet specs

▶ Two Kinds of Analysis

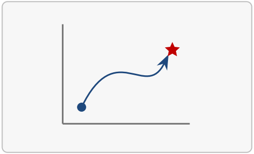
- **Quantitative**: what is the response to this input?
- **Qualitative**: is it stable? can the input steer every state? does the output reveal the state? → **source of design insight**



what response?



stable?



controllable?

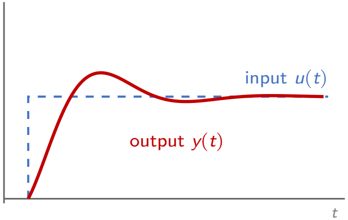
Response and Stability — Two Analysis Questions

CONCEPT Response (quantitative)

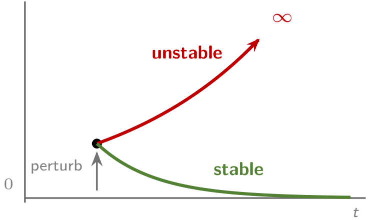
- **Given**: a model *and* an input signal $u(t)$
- **Ask**: which output trajectory $y(t)$ follows? $\rightarrow$ answer is **a curve**

CONCEPT Stability (qualitative)

- **Given**: model and a *nudge* off equilibrium
- **Ask**: does the state come back? $\rightarrow$ answer is **yes or no**



model + input
$\Rightarrow$ one definite curve



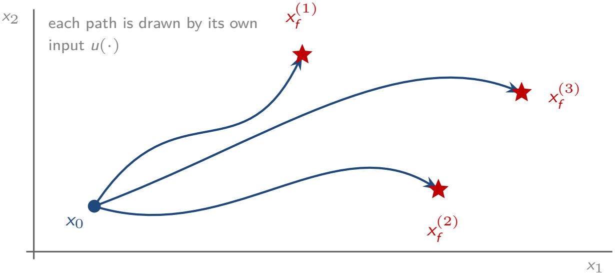
same nudge
$\Rightarrow$ return, or run away

Quantitative asks *how much*; qualitative asks *whether* — and design decisions rest on the second.

Controllability — Can Input Steer State Anywhere?

CONCEPT Controllability (qualitative)

- **Given**: model, a start x_0, and *any* target x_f in the state space
- **Ask**: is there an input $u(\cdot)$ driving x_0 to x_f in **finite time** — for *every* target? $\rightarrow$ answer is **yes or no**



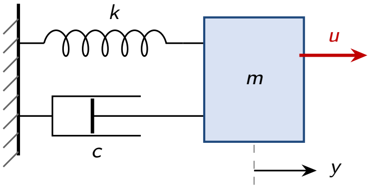
every target reachable in finite time $\Rightarrow$ controllable

This course follows one qualitative property — **controllability** —
asked repeatedly under **progressively weaker knowledge** of the model.

The Classical World — an A Whose Numbers We Know

- Classical control grew around a **single plant**: parameters measured on a test bench, identified from data, printed on a datasheet
- The world of Part I: $\dot{x} = Ax + Bu, \quad y = Cx$ — **every number in A trusted**
- Examples: mass–spring–damper, DC motor — m, c, k, R, L all known

EXAMPLE Mass–Spring–Damper



y measured from equilibrium

$$m\ddot{y} + c\dot{y} + ky = u$$

state $x = (y,\ \dot{y})^\top$

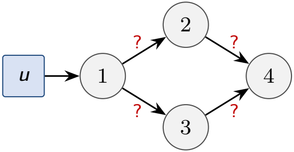
$$A = \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 1/m \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Measure m, c, k once on the bench and **every entry of A becomes a trusted number** — this is exactly the world Part I assumes.

Networked Systems — Knowledge Inverted

- Single plant: measure m, c, k once $\rightarrow$ A **is all numbers**. Networks **invert** this
- **Structure is reliable** — wiring diagram, communication topology, regulatory map
- **Coupling strengths are not** — they drift, resist measurement at scale

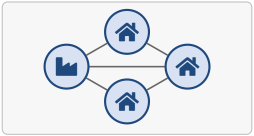


$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ * & 0 & 0 & 0 \\ 0 & * & * & 0 \end{bmatrix} \qquad B = \begin{bmatrix} * \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

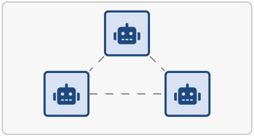
edges known · weights ?

$*$ unknown nonzero · 0 no edge, known exactly

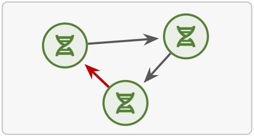
same information, two languages — the dictionary this book is built on



power grid



robot team

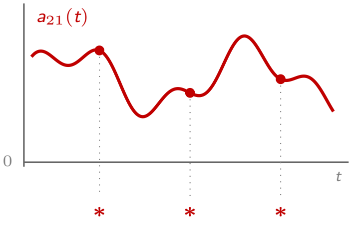


gene regulatory network

Weights Drift — Numerical Verdicts Expire

CONCEPT Values Move, Pattern Stays

- Even the "constants" are not constant: $a_{ij} = a_{ij}(t)$ with temperature, load, operating point
- So a rank verdict computed from *today's* numbers **expires tomorrow** — yet the edge is still there



value moves · pattern entry * never does

What Can Still Be Decided

- Holds for **every** value $\rightarrow$ drift cannot break it (*strong structural,* Ch. 5)
- Holds for **almost all** values $\rightarrow$ survives the drift (*structural,* Ch. 5)
- Stability, same idea: connected graph + positive weights $\rightarrow$ states **agree**, whatever the numbers do

[Premise of Part II] If the numbers in A cannot be trusted but the **zero/non-zero pattern** can, what remains of controllability?

Teaser — One Number Flips the Verdict

EXAMPLE Same graph, opposite verdicts (formally in week 3)

- Two systems of **identical pattern**: same nodes, same edges — **one** edge weight differs

$$A_\circ = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \text{ (uncontrollable)} \qquad A_\bullet = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} \text{ (controllable)}$$

- On a lab plant: a curiosity resolved by measurement
- On a network: **the weights are exactly what we do not know** $\rightarrow$ a numerical verdict certifies nothing

What can the pattern alone certify? — this question opens all of Part II. The answers are combinatorial: **read off a graph**.

Three Parts, One Chain of Questions

Part I Foundations (Ch. 1–3) — state-space models and **their graphs**, Kalman rank

↓ what can the pattern certify?

Part II Analysis (Ch. 4–8) — trust the pattern only: SC · SSC, subspaces, graph criteria, fixed nodes

↓ where should the inputs go?

Part III Synthesis (Ch. 9–10) — how many inputs, and where: minimum & optimal placement

- Each chapter opens by answering **the question the previous one left**
- See the one-page roadmap figure in the Preface of the lecture notes

Two Symmetries Running Through the Book

- **Symmetry 1 — SC vs SSC**
  - **Structural (SC)**: holds for *almost all* weights (generic)
  - **Strong structural (SSC)**: holds for *all* weights (worst case)
- **Symmetry 2 — Algebra vs Graph**
  - Every condition stated in **both languages**: determinant · rank $\leftrightarrow$ stems · matchings · zero forcing
- **Example Philosophy**
  - Mostly **4–5 nodes**, verifiable by hand
  - Each concept shown on the **smallest graph** that exhibits it

Prerequisites and Next Week

- **Prerequisites (all reviewed in week 2)**
  - Undergraduate control (transfer functions, basic state space)
  - Linear algebra (eigenvalues, rank) · basic programming (MATLAB)
  - Graph theory is **not assumed** — introduced within the course
- **Next Week (Ch. 2 — Foundations)**
  - State-space models and the **graph of a linear system** $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ — the book's declared viewpoint
  - Linear-algebra toolkit · solutions and stability · Laplacian consensus dynamics

[Line of the day] "Every linear system is already a graph.
Part II merely **erases the numbers** on the edges."